something truly surprising lurking within it.

Zeh had been puzzling over a problem in nuclear physics, one in which an atomic nucleus was in a Schrödinger's cat–like state of superposition, pointing in a multitude of directions at once. Meanwhile, the protons and neutrons within the nucleus were highly entangled with one another, so finding the position of just one of them would determine the positions of all the rest. "This made me think," Zeh recalled. "I said, let's assume that the universe is a closed system, like a nucleus. It was for me a very important step." Zeh didn't think that the universe was literally a single atomic nucleus. But he realized that the general idea—a system in a superposition, with its components strongly entangled—could explain how measurement works in quantum physics, without resorting to any of the tricks the Copenhagen interpretation used, like wave function collapse or a split between the physics of the small and the large. Treat a measurement device as a quantum system, and the act of measurement as a normal physical interaction, and quantum physics says that the measurement device will become strongly entangled with the thing it's measuring—and the overall system of measuring-device-and-thing-being-measured will be in a Schrödinger's cat state. But, Zeh realized, it went beyond that: the measuring device interacts with the experimenter, and everything else in the room, and eventually the entire universe—so when a small quantum system interacts strongly with a large object, ultimately, the entire universe ends up like Schrödinger's cat, splitting into dead-cat and alive-cat "branches." And the inhabitants of each branch of the universe only see one outcome: the dead cat or the living cat, depending on which branch they're in. But the wave function never collapses, and the different branches of the universe are extraordinarily unlikely to interact. "If you make a measurement, you get an entanglement between the system and the apparatus and the observer," Zeh said. "The observer sees only one component [of the Schrödinger's cat state] and not the superposition of all the others. So, that solves the measurement problem." Zeh had unknowingly reinvented Everett's many-worlds interpretation from scratch—and, along the way, he had also developed a mathematically sophisticated account of the interactions between small quantum systems, like atoms, and the relatively large quantum objects around them, like rocks and trees and measurement devices. This, in turn, explained why the different branches of the universal